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Cu NMR and NQR Studies of High-T_c Superconductor $La_{2-x}Sr_xCuO_4$

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Results of extensive Cu NMR and NQR studies in the superconducting $La_{2-x}Sr_xCuO_4$ with $x=0.075\sim0.24$ have been presented. The rapid decrease of the nuclear spin-lattice relaxation rate, $1/T_1$, and the spin Knight shift below T_c confirm microscopically the existence of the superconducting phase with lower T_c than 38 K for $0.10\leqq x\leqq 0.24$. For $x=0.10$, magnetic anomalies emerge at low-T in some small parts of the sample, *e.g.* T-independent T_1 and broadening of spectrum, having some relevance to the magnetic order in $La_{2-x}Ba_xCuO_4$ with $x=0.125$. For $x>0.15$, the $T_1T=$const. behavior and the existence of the residual spin Knight shift far below T_c have revealed that the superconductivity is of a gapless type with a finite density of states at the Fermi level caused by the mixture of different structures, i.e. tetragonal and orthorhombic.

[$La_{2-x}Sr_xCuO_4$, high-T_c superconductivity, Cu NMR and NQR, d-wave, gapless superconductivity, antiferromagnetic spin correlation, structural instability]

§1. Introduction

The neutron diffraction[1] and nuclear magnetic resonance (NMR) experiments[2,3] have demonstrated that there remains a short-range antiferromagnetic (AF) spin correlation even in the normal state of the superconducting $La_{2-x}Sr_xCuO_4$ (LSCO). NMR and nuclear quadrupole resonance (NQR) measurements have an advantage in extracting information on the electronic state at each atomic site. So far, there are many NMR and NQR works on each atomic site in high-T_c superconductors.[4] In the normal state, the characteristic features are that $(T_1T)^{-1}$ of ^{89}Y[5] and ^{17}O[6] in $YBa_2Cu_3O_7$ ($YBCO_7$) are nearly T-independent, while that of ^{63}Cu decreases with increasing temperature.[4] At the present, the different T-dependences of $(T_1T)^{-1}$ between ^{63}Cu and ^{17}O, ^{89}Y are understood as due to the AF d-spin fluctuations among Cu spins, which is canceled out geometrically at O(2, 3) and Y sites.[6,7] Hence the spin fluctuations with the wave vector around $q=Q$ at the zone boundary, $Q=(\pi/a, \pi/a)$, is almost filtered at O and Y sites, so that their nuclear spins see only the d-spin fluctuations with wave vectors around $q=0$ at the zone center. In the superconducting state, $1/T_1$ decreases neither with the BCS type enhancement nor exponential dependence, which are seen commonly in high-T_c cuprate.

In the previous papers,[2,3] from systematic measurements on T- and Sr-content dependence of $1/T_1$, it has been shown that $(T_1T)^{-1}$ of ^{63}Cu in the normal state follows a characteristic T-dependence of $C/(T+\theta)$ associated with the Curie-Weiss law of the staggered susceptibility, $\chi_Q(T)$. The important finding is that the Weiss temperature, θ, decreases linearly with decreasing Sr content, being 0 K around $x=0.05$ where T_c tends to be 0 K. Apparently, the Cu relaxation rate of LSCO is largely enhanced in comparison with that of $YBCO_7$, reflecting the large contribution near the zone boundary.[3] The recent neutron experiments have shown that the wave number dependence of the dynamical response function is strongly peaked around incommensurate wave vectors and becomes sharper with lowering temperature, which are responsible for the Curie-Weiss behavior of $\chi_Q(T)$.[8]

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In the superconducting state of LSCO, we have reported that $1/T_1$ exhibits a different behavior from those for other systems. Namely, $1/T_1$ for the compounds with less than $x=0.15$ tends to be saturated after the rapid decrease below T_c.[3] The saturation of $1/T_1$ at low-T does not originate from the fluctuations of impurity spins, but is dominated by the spin fluctuations characteristic for the lightly-doped region less than $x=0.15$.[3]

Takagi *et al.*[9] have reported that the bulk superconducting phase is confined to a much narrower composition region than believed.[10] Recently, a systematic NMR study has, however, assured that there presents a superconducting phase with $T_c=18$ K in the compound with $x=0.24$.[11] A rapid decrease of the spin Knight shift is clear below $T_c=18$ K.[11] Furthermore, the $T_1T=$const. relation and the presence of the residual spin shift at low-T have provided a clear signature that the superconductivity in the Sr content region exceeding $x=0.15$ is of a gapless type similar to the results of Zn-doped $YBa_2Cu_3O_7$.[11,12]

The time is ripe to present extensive results on the magnetic and superconducting properties in $La_{2-x}Sr_xCuO_4$ studied by Cu NQR and NMR synthetically.

§2. Experimental Procedures

The samples of $La_{2-x}Sr_xCuO_4$ used in the present work were prepared by solid state reaction following the procedure reported previously.[13,14] In order to reduce the Meissner shielding effect below T_c, the sample was well ground, so that each grain was smaller than 20 μm in diameter. T_c's were determined for respective x to be 20($x=0.075$), 27(0.085), 30(0.10), 36(0.13), 38(0.15), 29(0.20), 24(0.22), and 18 K(0.24) by ac susceptibility measurement. Here T_c is defined as the temperature at which the diamagnetization appears.

The ^{63}Cu NQR experiment was performed in zero field using a phase-coherent pulsed NMR spectrometer. The NQR spectra of Cu in LSCO at 4.2 and 77 K were obtained by measuring the spin echo signal as a function of frequency point by point. T_1 of ^{63}Cu was measured with the saturation recovery method in zero field and in the T-range of 1.3 ~ 300 K. T_1 is determined from the relaxation function, $R(t)$, of $R(t)=[M(\infty)-M(t)]/M(\infty)=\exp(-3t/T_1)$ plotted against time t, where $M(t)$ is the nuclear magnetization at time, t, after the saturating pulses. In high-T region, $R(t)$ is well fitted by a simple exponential form, while in the low-T region, there appears a multi-exponential behavior. The distribution of T_1 in this system is considered to be due to the inevitable effect caused by the random mixture of La and Sr. When the relaxation behavior is not simple exponential, there remains an ambiguity in determining T_1. In order to see the overall T-dependence of T_1 over the wide T-range, we adopt the value tentatively obtained from the slope where $R(t)$ decreases from 100 to 37% (down to 1/e) of the initial value (T_{1S}), from 30 to 10% (T_1), and from 10 to 5% (T_{1L}), respectively.

The Knight shift of ^{63}Cu was measured on the samples with the c-axis aligned along a magnetic field of 11 T at room-T. The ^{63}Cu NMR spectrum in the T-range of 4.2 ~ 260 K was obtained by sweeping the magnetic field with the use of a superconducting magnet (12 T at 4.2 K) at a frequency of 125.1 MHz.

§3. Results and Discussions

3.1 Cu NQR spectrum

Figure 1 shows the Cu NQR spectra at 4.2 K for $x=0.10 \sim 0.30$ in $La_{2-x}Sr_xCuO_4$. Cu has two isotopes ^{63}Cu and ^{65}Cu with the nuclear spin $I=3/2$. The central peak of each spectrum corresponds to ^{63}Cu. The peak on the lower frequency side to the central peak corresponds to ^{65}Cu. We call the set of these two peaks of the ^{63}Cu and ^{65}Cu NQR lines set A. The full width at half maximum of the intensity, FWHM, at 4.2 K increases gradually upon Sr doping. For example, FWHM for $x=0.15$ is about 2.2 MHz, which is by one order of magnitude larger than about 0.22 MHz for $YBa_2Cu_3O_7$.[12] This large width is expected to be due to the random distribution of Sr content in LSCO. The reliable estimation for the T-dependence of the NQR frequency, ν_Q, is difficult because of the large width and the poor signal to noise ratio. The T-variation of ν_Q for $x=0.10 \sim 0.15$ is less than 0.2 MHz between 4.2 and 77 K.[3]

As seen in Fig. 1, there exists another ^{63}Cu NQR line on the high frequency side around

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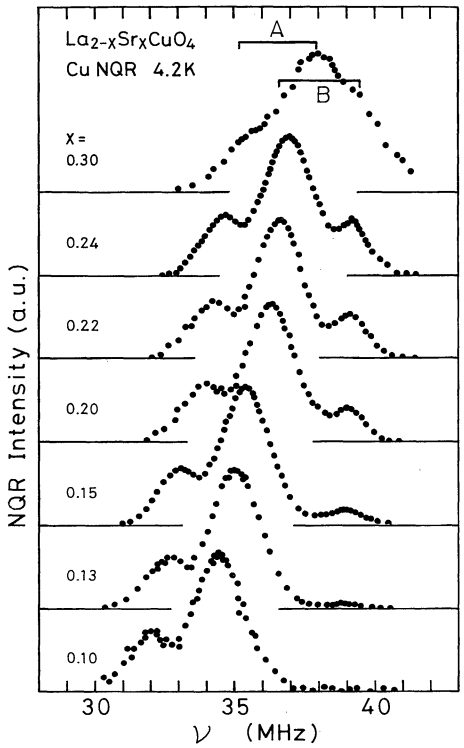


Fig. 1. Cu NQR spectra for $x=0.10\sim0.30$ in $La_{2-x}Sr_xCuO_4$ at 4.2 K in zero field. Here, A and B indicate the sets of ^{63}Cu and ^{65}Cu NQR lines.

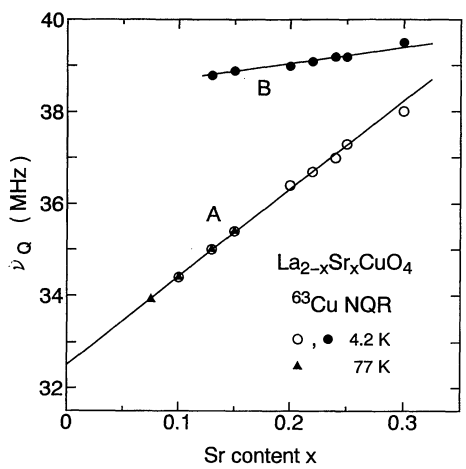


Fig. 2. x-dependence of the NQR frequency, ν_Q, of ^{63}Cu in $La_{2-x}Sr_xCuO_4$ at 4.2 and 77 K[3] in zero field. Here A and B indicate the ν_Q of ^{63}Cu for set A and B as shown in Fig. 1, respectively.

39 MHz (set B) as reported previously.[14,15] The ^{63}Cu NQR line of set B does not exist for $x=0.10$. The intensity of set B increases upon x from around $x=0.12$. Next, ν_Q for ^{63}Cu of set A and B are shown against Sr content, x, in Fig. 2. ν_Q for ^{63}Cu of set A increases linearly from 33.9 to 38 MHz with increasing x from 0.075 to 0.30 at 4.2 and 77 K.[3] On the other hand, ν_Q for ^{63}Cu of set B does not increase largely upon x, which is linked to ν_Q in non-superconducting samples[15] with x more than 0.30 transformed from the orthorhombic to the tetragonal structure. Therefore, the structure for Cu with the NQR line of set B may be like tetragonal. It may be possible that the structural instability is also associated with the presence of the oxygen deficiencies. We consider that the structural inhomogeneity is the origin of the gapless superconductivity in the Sr content region exceeding $x=0.15$ as discussed later.

3.2 *Normal state*

(1) T_1 of ^{63}Cu NQR

T_1 was mainly measured at ν_Q of ^{63}Cu in the set A (central peak of each spectrum) as shown in Fig. 1. Figure 3 shows the overall T-dependence of $1/T_1$ of ^{63}Cu for set A for $x=0.075\sim0.24$ plotted in a linear scale, together with that of $YBCO_7$ ($T_c=92$ K).[16] Some data have been reported previously.[2,3,11] $1/T_1$ for $x=0.075$ could only be measured above 77 K because of the poor signal to noise ratio due to the quite short spin echo decay time, T_2. As seen in the figure, $1/T_1$ is significantly enhanced as Sr content decreases, and the T-dependence of $1/T_1$ becomes weaker with increasing temperature.

Next, the T-dependence of $(T_1T)^{-1}$ is shown in Fig. 4. Near T_c, $1/T_1$ approaches to follow a linear T-dependence, especially for $x=0.20$ and 0.24. $(T_1T)^{-1}$ is largely enhanced as compared to that for the Cu metal [$(T_1T)^{-1}\sim1$ sec^{-1} K^{-1}] and increases extremely with decreasing temperature and Sr content. This strong T-dependence of $1/T_1T$ is due to the T-dependence of the staggered susceptibility, $\chi_Q(T)$. As discussed so far, the antiferromagnetic (AF) spin correlation plays a central role for the relaxation mechanism. As seen in Fig. 5, the Curie-Weiss law of $1/T_1T$, namely

$$(T_1T)^{-1}\sim C/(T+\theta), \qquad (1)$$

is confirmed for all the samples above θ K.

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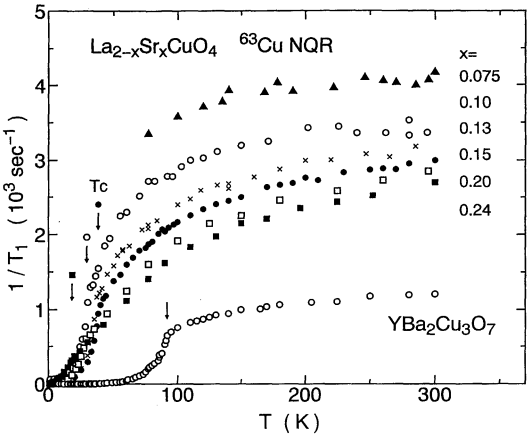


Fig. 3. T-dependence of $1/T_1(=2W)$ of ^{63}Cu for $x=0.075(\blacktriangle)$, $0.10(\circ)$, $0.13(\times)$, $0.15(\bullet)$, $0.20(\square)$, and $0.24(\blacksquare)$ in $La_{2-x}Sr_xCuO_4$ plotted in a linear scale, together with the data of $YBa_2Cu_3O_7(\circ)$.[16]

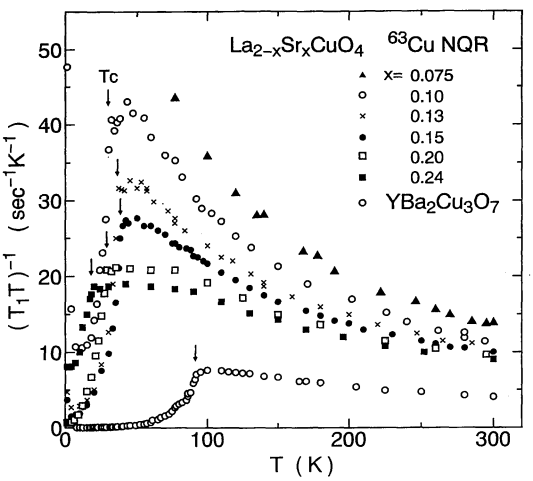


Fig. 4. T-dependence of $(T_1T)^{-1}(=2W/T)$ of ^{63}Cu for $x=0.075(\blacktriangle)$, $0.10(\circ)$, $0.13(\times)$, $0.15(\bullet)$, $0.20(\square)$, and $0.24(\blacksquare)$ in $La_{2-x}Sr_xCuO_4$ plotted in a linear scale, together with the data of $YBa_2Cu_3O_7(\circ)$.[16]

There have been many theoretical approaches for the Cu relaxation result,[17-21] *e.g.* the phenomenological AF fermi liquid model,[18] the SCR theory of spin fluctuations in a nearly AF two dimensional metal from a weak correlation limit[19] and the t-J model from the strong correlation limit.[21] Moriya *et al.* have extended the SCR theory of a weakly or nearly AF metal to the two dimensional system. The T-linear increase of the resistivity has been interpreted reasonably.[19] In particular, the T-dependence of T_1T in Fig. 5, which is inversely proportional to $\chi_Q(T)$, is quite similar to the calculated results of $1/\chi_Q(T)$. On the other hand, Kohno and Yamada derived rigorous expressions for the specific heat, the spin susceptibility, T_1, and the T^2 term of the electrical resistivity by a microscopic treatment consisting of d- and p-electrons based on a Fermi liquid picture.[18] Assuming large AF spin fluctuations and using our experimental finding that $(T_1T)^{-1}$ follows the Curie-Weiss law of $C/(T+\theta)$, they derived an expression of the resistivity as

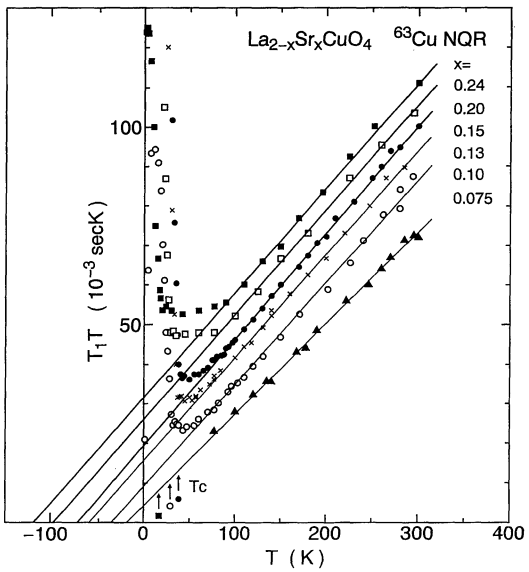


Fig. 5. T-dependence of $T_1T(=T/2W)$ of ^{63}Cu for $x=0.075(\blacktriangle)$, $0.10(\circ)$, $0.13(\times)$, $0.15(\bullet)$, $0.20(\square)$, and $0.24(\blacksquare)$ in $La_{2-x}Sr_xCuO_4$.

$$R\sim\chi_Q(T)T^2\sim T^2/(T+\theta)\quad(T\geqq\theta). \tag{2}$$

From this result, we see that the resistivity is proportional to T^2 with the coefficient enhanced by the $\chi_Q(T)$ at low-T far below θ and approaches T-linear like dependence at high-T far above θ. This appears to be the simplest interpretation for T-linear resistivity. For LSCO, since θ is almost comparable to T_c as Sr content is decreased, it is expected that T-linear resistivity is observed in the wide T-range above T_c, which is in accord with the experiment. The experimental results such as the T-dependence of $1/T_1$ and the resistivity, which were considered to be unconventional,

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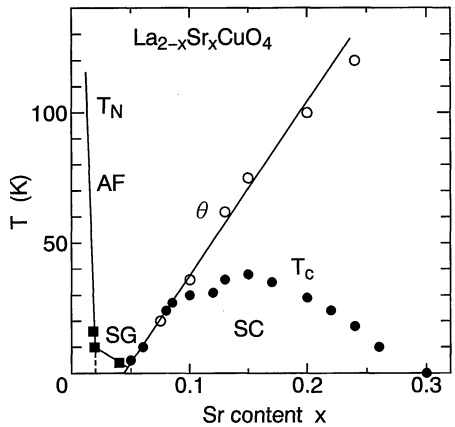


Fig. 6. θ(○) plotted against x, together with T_N(■) and T_c(●). θ is extracted by fitting the (T_1T) data to $(T+\theta)/C$ in Fig. 5. The solid line is a guide to eyes.

are well understood from the approaches based on the Fermi liquid picture in which the large AF spin fluctuations are taken into consideration.

θ extracted from the T_1T vs T plot in Fig. 5 is plotted against the Sr content, x, in Fig. 6, together with T_c and the Néel temperature, T_N. In the figure, the solid line is a guide to eyes. θ's for $x \leqq 0.20$ are those reported previously.[2,3,11] θ is linearly extrapolated to 0 K at around $x=0.05$ with $T_c=0$ K. There is a notable interrelation between T_c and θ, namely both decrease with decreasing Sr content, being $T_c=0$ K at $\theta=0$ K. It is remarkable that θ for $x=0.24$ is comparable to that in $YBa_2Cu_3O_7$ which is considered to have an optimum hole content giving a peak of T_c when T_c is plotted against the hole content as discussed by Zheng *et al.*[22] According to the results by Zheng *et al.*, the local hole density at each Cu and oxygen site is still located in the smaller region than the local hole content for $YBa_2Cu_3O_7$. Thus, we suppose that even the LSCO with x larger than 0.15 is not still in the over-doped region although T_c is lower than the maximum value of 38 K. T_c is, we infer, inevitably suppressed by the structural inhomogeneity as mentioned in §3.1, which causes the gapless superconductivity.

(2) Knight shift

The principal axis of the electric field gradient at the Cu site is almost parallel to the crystallographic c-axis because of the $3d_{x^2-y^2}$ character of the wave function. In order to determine the quadrupolar and Knight shift separately, we took series of spectra at several different frequencies in the region of 95.1 ~ 125.1 MHz. According to the second order perturbation theory, the resonance frequency, ν_Q, for the central transition spectrum with $H \perp c$ is given by

$$\nu_0=(1+K)\gamma_N H_{res}+\frac{\nu_Q^2[I(I+1)-3/4]}{16(1+K)\gamma_N H_{res}}, \tag{3}$$

where the first term involves the Knight shift and the second term is the second order quadrupolar shift. Therefore, we expect a linear relation between $\Delta\nu/(\gamma_N H_{res})$ and $(\gamma_N H_{res})^{-2}$ as

$$\frac{\Delta\nu}{\gamma_N H_{res}}=K+\frac{3\nu_Q^2}{16(1+K)(\gamma_N H_{res})^2}. \tag{4}$$

This relation is confirmed experimentally as seen in Fig. 7. Actually, the ν_Q's extracted from the slope of lines at 4.2 K in the figure are 37.2(NQR=37), 35.9(36.4), 34.9(35.4) and 34.2(34.4) MHz for $x=0.24$, 0.20, 0.15 and 0.10, respectively, being almost consistent with the results of NQR experiment. The intercepts of these lines, i.e. the values at $(\gamma_N H_{res})^{-2}=0$, give the Knight shift.

In general, the Knight shift, K, may be expressed as

$$K_\alpha(T)=K_{spin\,\alpha}(T)+K_{orb\,\alpha}, \quad (\alpha=/\!/,\ \perp) \tag{5}$$

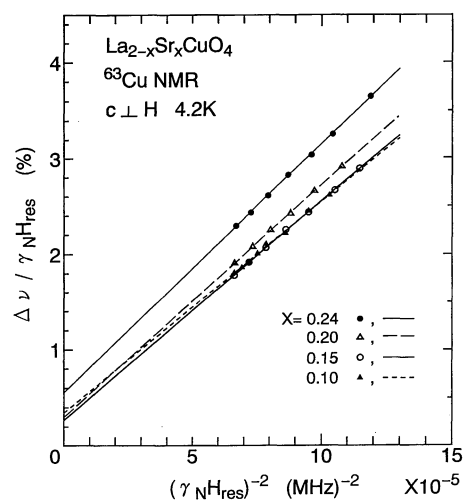


Fig. 7. $\Delta\nu/\gamma_N H_{res}$ vs $(\gamma_N H_{res})^{-2}$ plot in the Sr content region of $x=0.10 \sim 0.24$ in $La_{2-x}Sr_xCuO_4$.

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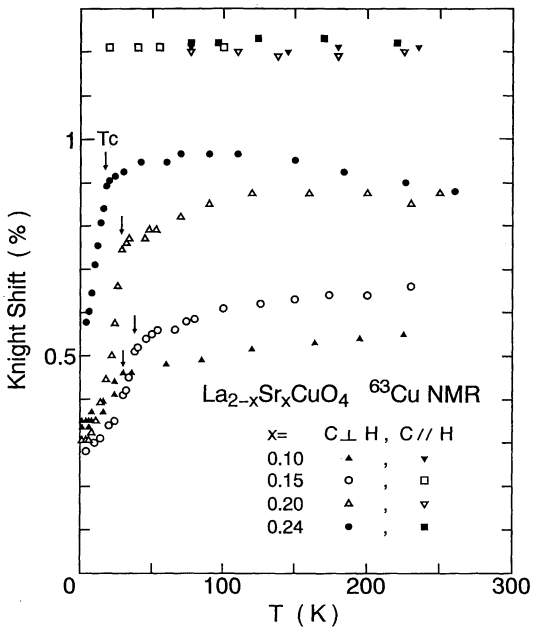


Fig. 8. T-dependence of the Knight shift parallel to the c-axis, $K_{/\!/}$, for $x=0.1$(▼), 0.15(□), 0.20(▽), and 0.24(■) and perpendicular to the c-axis, $K_{\perp}$, for $x=0.1$(▲), 0.15(○), 0.20(△), and 0.24(●) of ^{63}Cu in $La_{2-x}Sr_xCuO_4$, respectively.

where $K_{\rm spin}$ and $K_{\rm orb}$ indicate the spin and orbital Knight shift, respectively. $K_{{\rm spin}\,\alpha}(T)$ being T-dependent is coupled to the spin susceptibility, $\chi_{\rm spin}(T)$, as

$$K_{{\rm spin}\,\alpha}(T)=\frac{A_{\rm hf}}{N\mu_{\rm B}}\chi_{\rm spin}(T), \qquad (6)$$

where $A_{\rm hf}$, N, and $\mu_{\rm B}$ indicate the hyperfine field, the Avogadro's number and the Bohr magneton, respectively. Here $A_{\rm hf}$ in this system is described by the term of $(A_{/\!/,\perp}, +4B)$ based on the Mila-Rice model where $A_{/\!/,\perp}$ and B are the on-site hyperfine field parallel and perpendicular to the c-axis and the isotropic super-transferred hyperfine field, respectively.[7] Thus the measurements of NMR shifts yield information on the local magnetic susceptibility.

Figure 8 shows the T-dependence of the Knight shift parallel, $K_{/\!/}$, and perpendicular, $K_{\perp}$, to the c-axis for $x=0.1\sim0.24$ of ^{63}Cu in $La_{2-x}Sr_xCuO_4$, respectively. As seen in the figure, $K_{/\!/}$ of about 1.2% is x- and T-independent, being almost the same value as those in $YBCO_7$[23] and heavily-doped Tl(2201).[24] $K_{/\!/}$ is attributed to the orbital Knight shift because $K_{/\!/}$ does not change below T_c. Since $K_{/\!/}$ is x-independent, $K_{{\rm orb}\perp}$ is considered to be also x-independent. $K_{\rm orb}$ depends on the $3d$-hole density, crystal field splitting, and $3d$-orbital hyperfine coupling. The material-independent value of $K_{{\rm orb}/\!/}$ suggests that the above quantities do not change over whole Sr content. In fact, analysis of ν_Q of Cu by Zheng *et al.*[22] suggests that the hole density at Cu site does not change appreciably in high-T_c materials. $K_{{\rm orb}\perp}$ for $x=0.15$ is estimated to be 0.24% by extrapolating $K_{{\rm orb}\perp}(T)$ to $T=0$ in Fig. 8. Thus $K_{{\rm orb}\perp}$ for $x=0.10\sim0.24$ is also assumed to be 0.24% being the same value of $K_{\perp}$ at low-T far below T_c for $YBCO_7$[23] and Tl(2201).[24] Here we ignore the contribution of the diamagnetic shift perpendicular to the c-axis, because it is less than -0.03% at 4.2 K in the magnetic field of 11 T as shown from the proton NMR in $YBCO_7$.[25] Then as seen in Fig. 8, $K_{{\rm spin}\perp}(T)$ increases upon x as a whole with a weak T-dependence. This feature of the Knight shift of Cu is almost consistent with that of ^{17}O in CuO_2 planar site for $x=0.075$,[22] 0.15,[26] and 0.24[22] in $La_{2-x}Sr_xCuO_4$. The increase of $K_{\perp}$ upon hole doping means the increase of the uniform susceptibility, χ_o, at the zone center, $q=0$. It is pointed out from this result and the x-dependence of $1/T_1T$ that χ_o increases as χ_Q decreases upon hole doping.

3.3 *Superconducting state*

(1) Sr content region of $x\leqq0.15$

$1/T_1(=2W)$ of ^{63}Cu NQR in the set A as shown in Fig. 1 for $x=0.075\sim0.15$ in $La_{2-x}Sr_xCuO_4$ is plotted in a logarithmic scale in Fig. 9. T_{1S} and T_{1L} are also shown in the figure, respectively, to show the extent of the distribution of T_1 below T_c. For a comparison, the result for $YBa_2Cu_3O_7$[16] is also shown. $1/T_1$ of ^{63}Cu in LSCO is considerably larger than that in YBCO both above and below T_c. Although the distribution of T_1 becomes remarkable with decreasing temperature below T_c, all the components of $1/T_1$ for $x=0.10$ and 0.15 are confirmed to decrease below T_c.[3,11] In addition, the spin Knight shift perpendicular to the c-axis for $x=0.10$ and 0.15 also decreases below T_c as seen in Fig. 8. These results indicate that there is no normal part in the sample and no significant distribu-

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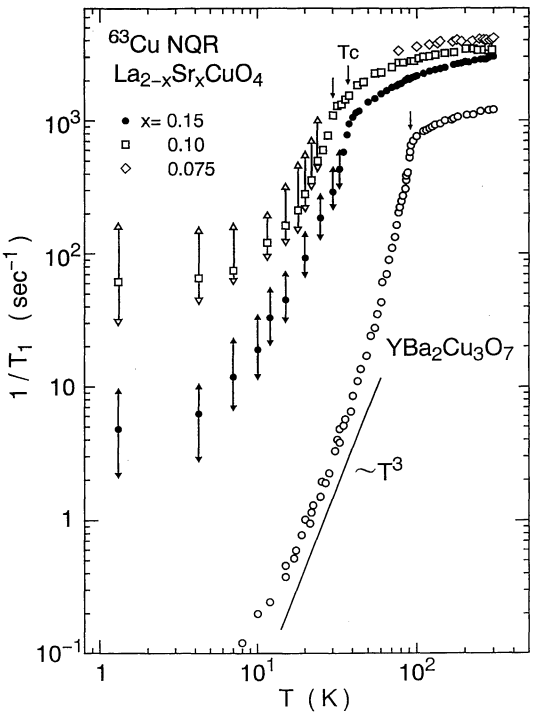


Fig. 9. T-dependence of $1/T_1(=2W)$ of ^{63}Cu for $x=0.075(\diamond)$, $0.10(\square)$, and $0.15(\bullet)$ in $La_{2-x}Sr_xCuO_4$ (LSCO) and that of $YBa_2Cu_3O_7(\circ)$[16] with $T_c=20$, 30, 38, and 92 K, respectively.[3,11] ○ and ● for LSCO represent $1/T_1$ evaluated from the slope where $R(t)=[M(\infty)-M(t)]/M(\infty)$ decreases from 30 to 10% of the initial value. △, ▲ and ▽, ▼ indicate $1/T_1$ obtained from the slope where $R(t)$ decreases from 100 to 37% and from 10 to 5%, respectively, i.e. T_{1S} and T_{1L}. Here $M(t)$ is the nuclear magnetization at time, t, after the saturating pulses.

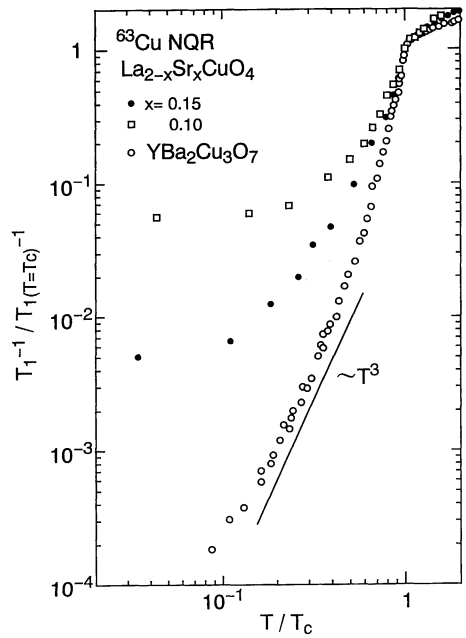


Fig. 10. Normalized $1/T_1$, i.e. $T_1^{-1}/T_{1(T=T_c)}^{-1}$, of ^{63}Cu for $x=0.10(\square)$, and $0.15(\bullet)$ in $La_{2-x}Sr_xCuO_4$ (LSCO) and that of $YBa_2Cu_3O_7(\circ)$[16] are plotted against T/T_c.

tion of T_c.

In order to emphasize the characteristic feature below T_c, the T-dependence of $1/T_1$ normalized by the value at T_c for $x=0.10$ and 0.15 in $La_{2-x}Sr_xCuO_4$ and $YBa_2Cu_3O_7$[16] is plotted against T/T_c in Fig. 10. As seen in the figure, the behavior of the rapid reduction of $1/T_1$ without the enhancement below T_c is independent of Sr content and T_c. This reduction behavior is a common feature not only observed in the Cu relaxation in the CuO_2 plane[27] but also in the relaxation of ^{17}O[6] and ^{89}Y[5] in $YBCO_7$. Accordingly, it is evident that the same mechanism works in the relaxation process just below T_c which cannot be interpreted by the conventional BCS model.[28] Here it should be emphasized that the terminology of BCS model is used in a usual sense that an isotropic energy gap openes at the Fermi level below T_c, which causes the enhancement of $1/T_1$ just below T_c. Furthermore, we have already found an important result that $1/T_1$ of LSCO deviates from a steep decrease and tends to be saturated at lower temperatures with further decreasing temperature as seen in Fig. 10.[3,11] The saturated value increases as Sr content is decreased, which cannot be attributed to trivial impurities.

Figure 11(a) and 11(b) show the T-dependence of the relaxation function, $R(t)$, for $x=0.10$ in zero (NQR) and external magnetic field, H_{ext}, (NMR), respectively. As seen in Fig. 11(a), most of the components of T_1 are shown to be T-independent at 1.3 ~ 4.2 K. On the other hand, in H_{ext}, the T_1=const. behavior disappears, instead $1/T_1$ decreases with decreasing temperature from 4.2 to 1.3 K. The broken line in Fig. 11(b) is the calculation for H_{ext} parallel to the c-axis by using the data obtained from NQR [the solid line in Fig. 11(a)]. Here the relation $R(t)=0.1\exp(-t/T_1)+0.9\exp(-6t/T_1)$ in NMR is used while

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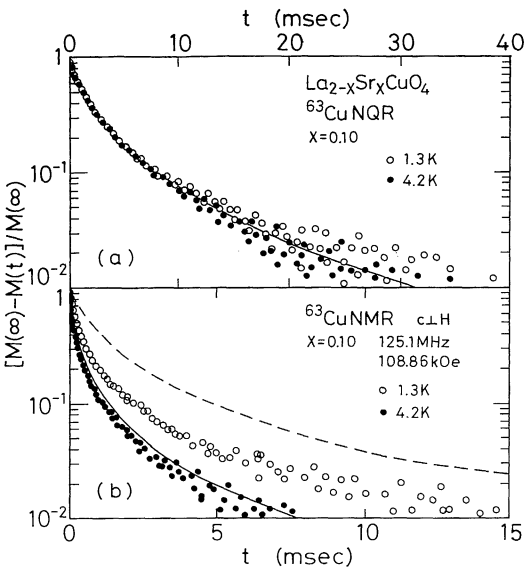


Fig. 11. Relaxation function, $R(t)=[M(\infty)-M(t)]/M(\infty)$, of ^{63}Cu plotted against the time, t, for $x=0.10$ in $La_{2-x}Sr_xCuO_4$ at 1.3(○) and 4.2 K (●) in zero field (NQR) and in the magnetic field of 108.86 kOe (NMR) perpendicular to the c-axis, $c\perp H$, respectively. Here, the broken line is the calculation for $R(t)$ parallel to the c-axis, $c/\!/H$, at 4.2 K by using the NQR data [solid line in Fig. 11(a)]. The solid line in Fig. 11(b) is the expected $R(t)$ for $c\perp H$ (see the text).

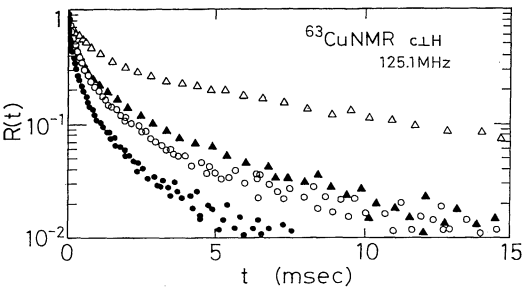


Fig. 12. Relaxation function, $R(t)=[M(\infty)-M(t)]/M(\infty)$, of ^{63}Cu plotted against the time, t, for $x=0.085$ in $La_{2-x}Sr_xCuO_4$ at 1.6(△) and 4.2 K (▲) in the magnetic field perpendicular to the c-axis, $c\perp H$. ○ and ● indicate $R(t)$ for $x=0.10$ at 1.3 and 4.2 K, respectively, i.e. those plotted in Fig. 11(b).

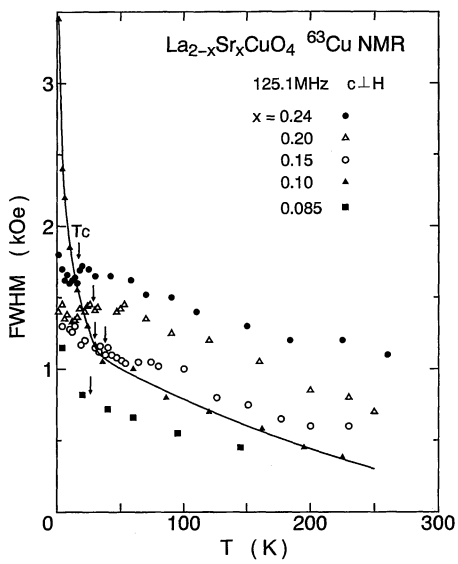


Fig. 13. Full width at half maximum of the intensity, FWHM, of ^{63}Cu NMR. Here, ■, ▲, ○, △, and ● represent the value of FWHM with $c\perp H$ for $x=0.085$, 0.10, 0.15, 0.20, and 0.24, respectively. The solid line for $x=0.10$ is a guide to eyes.

$R(t)=\exp(-3t/T_1)$ in NQR. Now at 77 K we measured the anisotropy of $T_{1/\!/}/T_{1\perp}$ to be 3.5. If we assume this relation to hold roughly at 4.2 K, then $R(t)$ for $H\perp c$ is expected as the solid line in Fig. 11(b). Thus, $1/T_1$ is considered to be suppressed by the external field at 1.3 K. The T-independent T_1 observed in $H_{ext}=0$ is considered to come from a part by a spin diffusion effect. The application of the field suppresses the diffusion. Thus most of the parts for $x=0.10$ have $1/T_1$ which decreases with decreasing temperature. This means that there exists a small region where $T_1=$const. holds when Sr content decreases. Figure 12 shows the relaxation function, $R(t)$, for $x=0.085$ in comparison with that for $x=0.10$. As seen in the figure, $1/T_1$ for $x=0.085$ at 4.2 K is smaller than those for $x=0.10$ at 4.2 and 1.3 K. Furthermore, the T-dependence for $x=0.085$ is very large. $1/T_1$ at 4.2 K is by four times larger than that at 1.6 K. This means the saturating behavior in $1/T_1$ appears remarkably for $0.085<x<0.15$.

Figure 13 shows the full width at half maximum of the intensity, FWHM, with $c\perp H$ of ^{63}Cu NMR for $x=0.085\sim0.24$. FWHM increases with decreasing temperature or increasing Sr content. As seen in Fig. 8, the Knight shift with $c\perp H$ in the normal state decreases for $x\leqq0.15$, while increases for $x\geqq0.20$. For each x, the distribution of K is expected to be associated with the distribution of Sr and La. Therefore, the T-dependence of K of respective Cu is different in each x, depending on the

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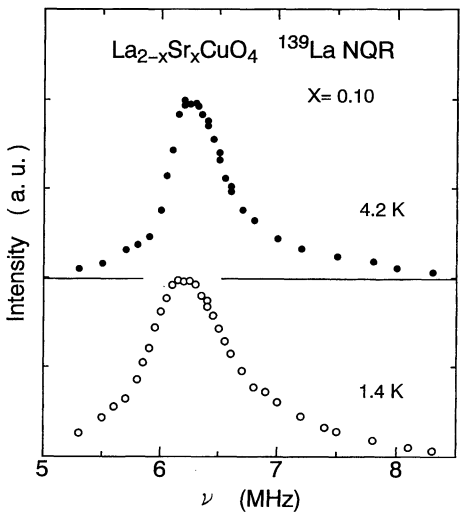


Fig. 14. ^{139}La NQR spectra for $\pm 1/2 \leftrightarrow \pm 3/2$ transition of $x=0.10$ at 4.2 and 1.4 K.

local environment. The increase of the line width seen commonly for the whole x is attributed to the distribution of K. Now, the FWHM for $x=0.10$ increases extremely at low-T in contrast to those for the other Sr content than $x=0.10$ even in the superconducting state, although the bulk susceptibility possesses no Curie like upturn. In addition, the line width of ^{139}La NQR spectrum for $\pm 1/2 \leftrightarrow \pm 3/2$ transition of $x=0.10$ increases with decreasing temperature from 4.2 to 1.4 K as seen in Fig. 14, which is ascribed to the internal field associated with the occurrence of a short-range magnetic order. The anomalies, i.e. the saturation of $1/T_1$ and the marked increases of the NMR line width observed far below T_c, may be attributed to a critical slowing down of the Cu spin fluctuations associated with a possible magnetic ordering in some small parts of the sample, which is similar to that reported at low-T around $x=0.12$ in $La_{2-x}Ba_xCuO_4$.[29] More detailed results will be presented elsewhere.

(2) Sr content region of $x \geqq 0.15$

(a) NQR

Figure 15 shows the T-dependence of $1/T_1(=2W)$ of ^{63}Cu in the set A as shown in Fig. 1 for $x=0.15 \sim 0.24$ in $La_{2-x}Sr_xCuO_4$ together with that of $YBa_2Cu_3O_7$[16] in zero field. In the normal state, the T-range where $T_1T=$const. like behavior holds above T_c ex-

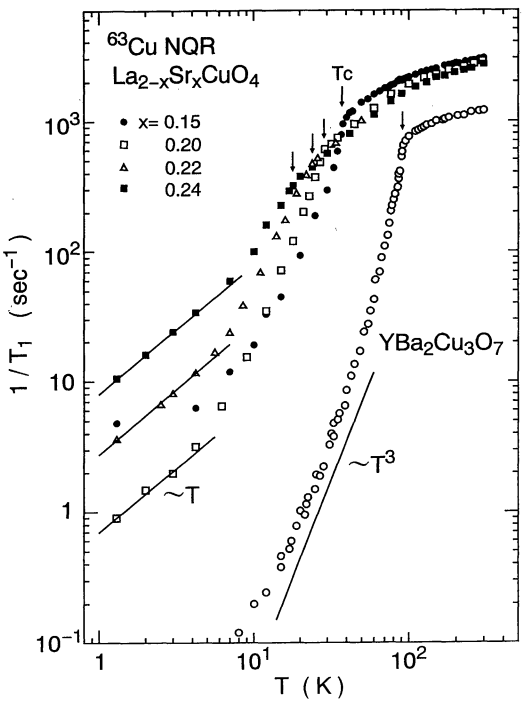


Fig. 15. T-dependence of $1/T_1(=2W)$ of ^{63}Cu for $x=0.15$(●), 0.20(□), 0.22(△), and 0.24(■) in $La_{2-x}Sr_xCuO_4$ together with that of $YBa_2Cu_3O_7$ (○)[16] in zero field.

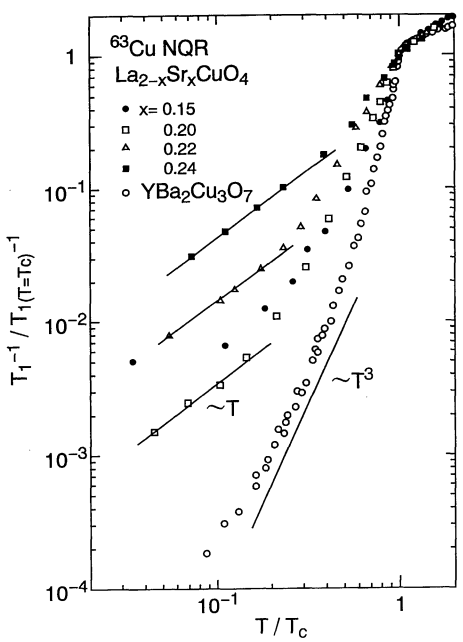


Fig. 16. Normalized $1/T_1(=2W)$, i.e. $T_1^{-1}/T_{1(T=T_c)}^{-1}$, of ^{63}Cu for $x=0.15 \sim 0.24$ in $La_{2-x}Sr_xCuO_4$ and that of $YBa_2Cu_3O_7$(○)[16] in zero field are plotted against T/T_c. The marks of T_1 for various compounds are the same as those in Fig. 15.

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tends toward higher Sr region as mentioned in §3.2(1).[3,11] In order to emphasize the characteristic feature below T_c as well as the lightly-doped region less than $x=0.15$, the T-dependence of $1/T_1$ normalized by the value at T_c is plotted against T/T_c in Fig. 16. Here marks are the same as in Fig. 15. In the superconducting state, the rapid decrease of $1/T_1$ below T_c without enhancement has not only been observed in the Sr content region with $x \leqq 0.15$, but also in the region with $x=0.20 \sim 0.24$ in $La_{2-x}Sr_xCuO_4$.[11] In addition, all the components of $1/T_1$ decrease rapidly below T_c over whole Sr content region. There is a doubt that the NQR line of set B is not from superconducting part because its structure may correspond to the non-superconducting tetragonal structure as mentioned in §3.1. If this is the case, it may be expected that $1/T_1$ is not decreased below T_c. However, $1/T_1$ of ^{63}Cu for set B of $x=0.15$[14] and 0.24 having about 1/2 times smaller value than that for set A, decreases below T_c, and the T-dependence is almost the same as that of ^{63}Cu for set A. Thus, Cu for set B is also considered to be in the superconducting state. As seen in Fig. 16, the reduction rate below T_c changes upon x, i.e. $1/T_1$ decreases less rapidly with increasing x in contrast to the case for $x \leqq 0.15$, indicating a smearing of the density of states by the pair breaking effect as will be discussed later. With further decreasing temperature, $1/T_1$ for $x>0.15$ deviates from the steep decrease and tends to be proportional to temperature at lower temperatures as Sr content increases from $x=0.20$ to 0.24.[11] Actually, the relaxation function, $R(t)$, plotted against the product of time, t, and temperature, T, for $x=0.22$ and 0.24 in the T-range of $1.3 \sim 4.2$ K and $1.3 \sim 7$ K, respectively, fall approximately on a curve within an experimental scattering as shown in Fig. 17. This situation is seen for the relaxation curves of $R(t)$ of ^{63}Cu for set B of $x=0.24$ plotted against the product of t and T in the T-range of $1.3 \sim 4.2$ K as shown in Fig. 18. These results indicate that all the components of T_1 of Cu follow $T_1T=$const. relation at low-T.

(b) NMR

The central transition of NMR spectra of ^{63}Cu for $x=0.24$ in $La_{2-x}Sr_xCuO_4$ at 4.2 and

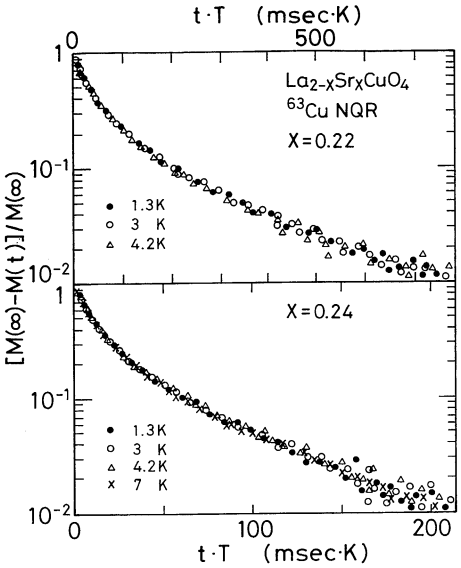


Fig. 17. Relaxation function, $R(t)=[M(\infty)-M(t)]/M(\infty)$ of ^{63}Cu in the T-range of $1.3 \sim 4.2$ K and in zero field for $x=0.22$ and of $1.3 \sim 7$ K for $x=0.24$ plotted against the product of time, t, and temperature, T.

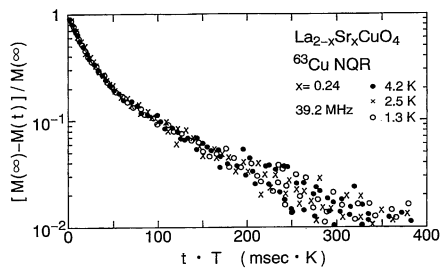


Fig. 18. Relaxation function, $R(t)=[M(\infty)-M(t)]/M(\infty)$ of ^{63}Cu for set B with $x=0.24$ as shown in Fig. 1 in the T-range of $1.3 \sim 4.2$ K and in zero field, plotted against the product of time, t, and temperature, T.

20 K and at the frequency of 125.1 MHz with the c-axis perpendicular to the magnetic field, $c \perp H$, are shown in Fig. 19. Here, the line shape of spectrum at 20 K is replotted on the spectrum at 4.2 K by open circles. As seen in the figure, the spectrum in the normal state at 20 K with the full width at half maximum of the intensity, FWHM, of 1.7 kOe is clearly shifted as a whole to that in the superconducting state at 4.2 K, and any trace indicating the normal part does not remain.[11] We conclude from this result and the NQR experimental results men-

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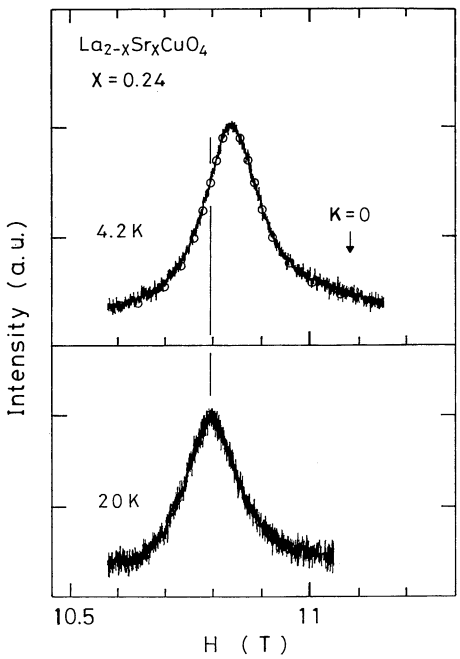


Fig. 19. Field swept ^{63}Cu NMR spectra for $x=0.24$ in $La_{2-x}Sr_xCuO_4$ with $c \perp H$ at 125.1 MHz at 4.2 and 20 K.[11] A repetition time between two pulses is 50 msec. Here, the shape at 20 K is replotted on the spectrum at 4.2 K by open circles. The intensity is normalized at the peak.

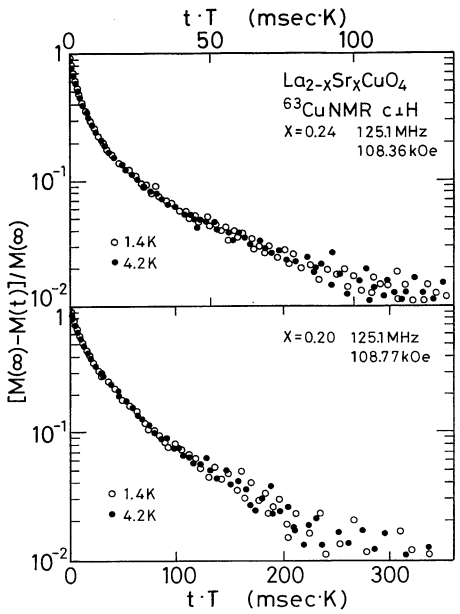


Fig. 20. Relaxation function, $R(t)$, of ^{63}Cu for $x=0.24$ and 0.20 with $c \perp H$ at 1.4 and 4.2 K, plotted against the product of time, t, and temperature, T.

tioned above that most of the Cu sites in the CuO_2 layer undergo a superconducting transition below T_c.

As seen in Fig. 8, $K_\perp$ for the whole x markedly decreases below T_c. The Knight shift is enhanced largely with increasing x from 0.15 to 0.24 at low-T far below T_c.[11] Since the orbital shift of $K_\perp$ has almost the same value of 0.24% over whole x as mentioned in §3.2(2) and the diamagnetic shift of $K_\perp$ is less than -0.03% at 4.2 K in the magnetic field of 11 T,[25] the residual Knight shift is ascribed to the spin shift.

Figure 20 shows the relaxation function, $R(t)$, of ^{63}Cu under the field with $c \perp H$ (NMR) at 1.4 and 4.2 K for $x=0.24$ and 0.20, plotted against the product of time, t, and temperature, T. $R(t)$ falls on a curve within the experimental errors, which means that all the components of T_1 follow again the $T_1T=$const. relation even in the *shifted signal* in Fig. 19. $1/T_1$ for each x in the field is almost the same as expected from $1/T_1$ in NQR measurement. The behavior of $T_1T=$const. and the presence of the residual spin shift make an evidence that a finite density of states is left even in the superconducting state, which means that the system is in a gapless superconducting regime. The detailed analysis will be presented in §4.

Recently, Oda *et al.* have proposed that the measured susceptibility, $dc\chi$, for LSCO cosists of the T-independent term, the spin susceptibility of Cu, $\chi_{spin}(T)$, and Curie term which appears above $x=0.18$ and increases with increasing x.[30] This T-dependence of $\chi_{spin}(T)$ extracted from $dc\chi$ is similar to the T-dependence of $K_\perp$ as shown in Fig. 8. They have proposed that the Curie term comes from the localized Cu spins, which destroy superconductivity. However, neither any remarkable increase of the width of NMR spectrum nor T-independent T_1 at low-T associated with the Curie term are seen upon Sr doping exceeding $x=0.15$ as seen in Figs. 13 and 15. This suggests that the Curie term does not come from the localized Cu spins. Here, we suggest a following possibility. A microscopical mixture of the different crystal structure destroys the AF correlation to produce clusters in which the

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short-range AF corelation still exists. The unquenched spins in the cluster including odd number of Cu spins could produce the Curie-Weiss behavior as a whole. Actually, the existence of an another phase with different structure, such as the tetragonal structure, is speculated from the existence of the different NQR spectrum in Fig. 1 as mentioned in §3.1. The decrease of T_c in the Sr content region over $x=0.15$ is not attributed to the localized Cu spins arising from the Curie term but to the pair breaking due to the inhomogeneity induced by the structural instability from orthorhombic to tetragonal phase.

§4. Analysis by the Gapless *d*-wave Model

In the superconducting state, $1/T_1$ of Cu in LSCO reveals a rapid decrease without the enhancement just below T_c. The rapid reduction of $1/T_1$ below T_c is known to be independent of the strength of the external field, the number of CuO_2 plane and the atomic sites, i.e. Cu,[27] O,[6] and Y,[5] in several high-T_c materials. So far, the Knight shift perpendicular to the *c*-axis, $K_\perp$, of Cu in LSCO is reduced below T_c even in the region of $0.15<x\leqq 0.24$ as well as the results of $1/T_1$, confirming the presence of superconducting phase with T_c less than the maximum T_c of 38 K from the microscopic point of view.

We show below that both of the T-dependence of T_1 and the Knight shift below T_c for $x=0.24$ and 0.20 can be interpreted by a gapless *d*-wave model. The steep decrease of $1/T_1$ just below T_c and its power-law like T-dependence at low-T in the high-T_c superconductors cannot be interpreted by an *s*-wave model, even if the quasiparticle damping is taken into account, in contrast to that of the Chevrel phase superconductor $Tl_{1.1}Mo_6Se_{7.5}$ where $1/T_1$ shows an exponential decrease at low-T although enhancement just below T_c is absent.[16] $1/T_1$ of Cu in LSCO decreases rapidly without the enhancement just below T_c, and a $T_1T=$const. relation and a residual spin Knight shift are seen at low-T far below T_c for $x>0.15$. The latter result provides an evidence that the superconducting state is of a gapless type.

We hence apply the model similar to that used in Zn-doped $YBa_2Cu_3O_7$[12] for the T-dependence of Knight shift and T_1, in which the gap zeros of lines at the Fermi surface originated from *d*-wave pairing and the residual density of states at the Fermi level associated with the depairing effect by non-magnetic impurity are taken into consideration. Actually, in order to interpret the physical quantities at low-T in heavy fermion superconductors (HFSC), this type of model was first argued by Schmitt-Rink, Miyake, and Varma[31] and Hirschfeld, Vollhardt, and Wolfle.[32] We adopt density of states based on a *d*-wave model with a gap function of approximately polar state, $C(\theta, \phi)=\Delta\cos\theta$, for simplicity as shown in Fig. 21. The functional forms are given by

$$N_s(E)=(\pi rN_0)/2=N_{res} \quad (E<r\Delta),$$
$$N_s(E)=(\pi N_0E)/(2\Delta) \quad (r\Delta<E<\Delta),$$

and

$$N_s(E)=(N_0E/\Delta)\sin^{-1}(\Delta/E) \quad (\Delta<E), \tag{7}$$

where $r=(2N_{res})/(\pi N_o)$. Here, the extent of the gapless is introduced only by a parameter, N_{res}, so the overall modification of the shape

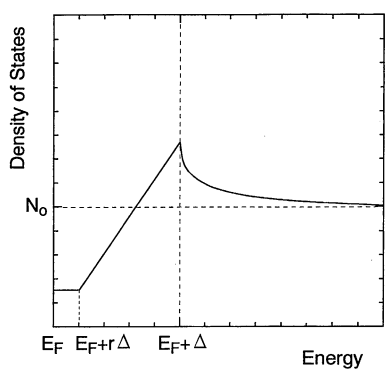


Fig. 21. Model density of states in which the gap zeros of lines at the Fermi surface originating from *d*-wave pairing and the residual density of states at the Fermi level associated with a pair breaking effect are taken into account. The functional forms are given by

$$N_s(E)=(\pi rN_o)/2=N_{res} \quad (E<r\Delta),$$
$$N_s(E)=(\pi N_oE)/(2\Delta) \quad (r\Delta<E<\Delta),$$

and

$$N_s(E)=(N_oE/\Delta)\sin^{-1}(\Delta/E) \quad (\Delta<E),$$

where $r=(2N_{res})/(\pi N_o)$.

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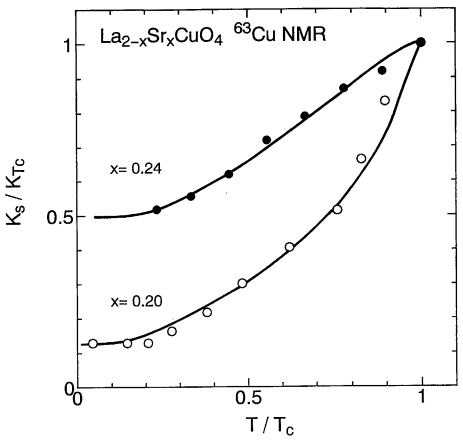


Fig. 22. T-dependence of K_s/K_{T_c} for $x=0.24$(●) and 0.20(○) plotted against T/T_c. The spin part of $K_\perp$, K_s, estimated by assuming the orbital part of $K_\perp$, K_{orb}, with 0.24%. K_{T_c} is the value at T_c. The solid lines are the calculation with $2\Delta=4.5k_BT_c$ and $r=0.32$ ($N_{res}/N_o=0.5$) for $x=0.24$ and with $2\Delta=8k_BT_c$ and $r=0.083$ ($N_{res}/N_o=0.13$) for $x=0.20$, respectively.

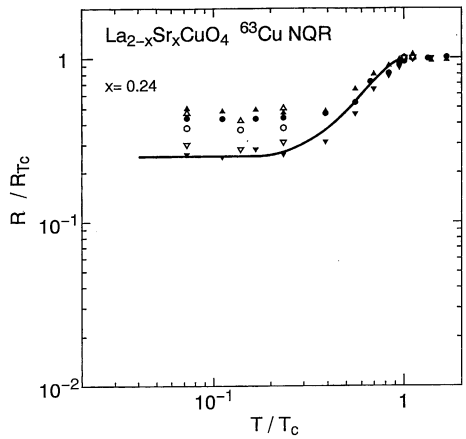


Fig. 23. Normalized $1/T_1T$ by the value at T_c, R/R_{T_c}, of ^{63}Cu for $x=0.24$ plotted agaist T/T_c. Here, ▲(△), ●(○), and ▼(▽) represent T_{1s}, T_1, and T_{1L} for set A(set B) as shown in Fig. 1, respectively. The solid lines are the calculation with the same parameters as those for the Knight shift in Fig. 22.

of $N_s(E)$ are neglected (broadening at $E=\Delta$ and the increase of the total area by introducing N_{res}). We try to interpret the T-dependence of the Knight shift and T_1 of Cu for $x=0.20$ and 0.24 by use of this model.

Figure 22 shows the T-dependence of K_s/K_{T_c} for $x=0.24$ and 0.20 plotted against T/T_c where the spin part of $K_\perp$, K_s, estimated by assuming the orbital part of $K_\perp$, $K_{orb}=0.24\%$, is normalized by the value at T_c. The solid lines are the calculation with $2\Delta=4.5k_BT_c$ and $r=0.32$ ($N_{res}/N_o=0.5$) for $x=0.24$ and with $2\Delta=8k_BT_c$ and $r=0.083$ ($N_{res}/N_o=0.13$) for $x=0.20$, respectively. Here the assumption of $K_{orb}=0.24\%$ in LSCO is supported by the facts as mentioned in §3.2(2) that the Knight shift parallel to the c-axis, $K_{/\!/}$, for $YBCO_7$[23] and LSCO in whole Sr content region is almost T-independent with the same value of about 1.2%, dominated by the orbital susceptibility. $K_{/\!/}$ for Tl(2201)[24] at low-T has the same value of 1.2%. In addition, the residual shift of $K_\perp$ dominated by the orbital susceptibility at low-T for $YBCO_7$[23] and Tl(2201)[24] have the same value of 0.24%. As seen in Fig. 22, K_s/K_{T_c} for $x=0.24$ and 0.20 are well fitted by the lines with $N_{res}/N_o=0.5$ and 0.13 and with $2\Delta=4.5$ and $8k_BT_c$, respectively. If the Knight shift can be interpreted by the gapless d-wave model, the results of T_1 should also be fitted by use of the same parameters as those for the Knight shift. Then we have tried to calculate T_1.

The normalized $1/T_1T$ by the value at T_c, i.e.

$$R/R_{T_c}=(1/T_1T)/(1/T_1T)_{T_c},$$

of ^{63}Cu for $x=0.24$ is plotted against T/T_c in Fig. 23. Here ▲(△), ●(○), and ▼(▽) represent T_{1s}, T_1, and T_{1L} for set A (set B) as shown in Fig. 1, respectively. As seen in Fig. 23, $1/T_1$ of Cu for two different sets are reproduced by the same gapless d-wave parameters as those for the Knight shift, and the Cu site for set B with tetragonal-like structure as mentioned in §3.1 is confirmed to be also in the superconducting state. In addition, the value of $1/T_1$ of Cu for set B is by 1/2 times smaller than that for set A in the normal state for $x=0.24$, which means the suppression of χ_Q for the Cu site in the tetragonal-like structure.

Next, R/R_{T_c} of ^{63}Cu for $x=0.20$ plotted against T/T_c is shown in Fig. 24. Here △, □, and ▽ represent T_{1s}, T_1, and T_{1L} for set A as shown in Fig. 1, respectively. The solid lines are the calculation by use of the same parameters as those of the Knight shift for $x=0.20$ as shown in Fig. 22. We consider that

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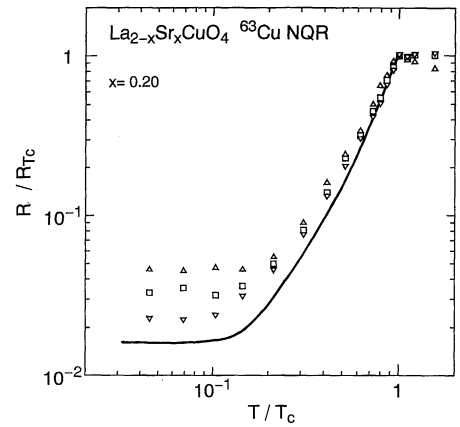


Fig. 24. Normalized $1/T_1T$ by the value at T_c, R/R_{T_c}, of ^{63}Cu for $x=0.20$ plotted against T/T_c. Here, △, □, and ▽ represent T_{1s}, T_1, and T_{1L}, respectively. The solid lines are the calculation with the same parameters as those for the Knight shift of $x=0.20$ as shown in Fig. 22.

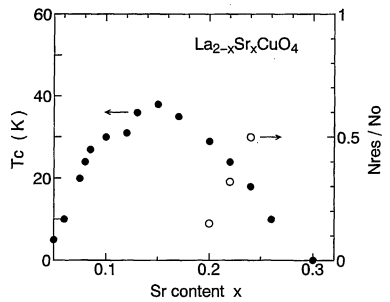


Fig. 25. N_{res}/N_o(○) and T_c(●) of $La_{2-x}Sr_xCuO_4$ plotted against x.

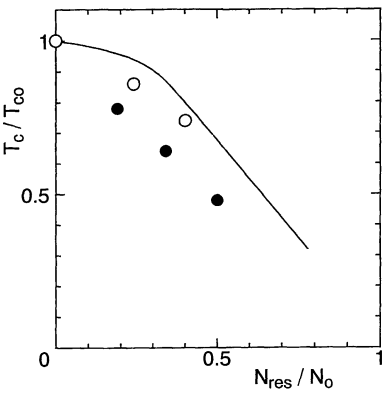


Fig. 26. T_c/T_{co} vs N_{res}/N_o plot. Here, ● and ○ represent the data for LSCO with $x \geqq 0.15$ and the Zn-doped $YBCO_7$,[12] respectively. T_{co} for LSCO and $YBCO_7$ are assumed as $T_{co}=38$ and 92 K, respectively. The solid line is the calculation based on the non-mgnetic impurity scattering in the unitarity limit by Schmit-Rink, Miyake, and Varma.[32]

T_1 for $x=0.20$ is also well interpreted by this model as well as for $x=0.24$, although there exists an experimental error for both of T_1 and Knight shift measurements.

As mentioned before, we neglected the modification of the density of states by the pair breaking effect for the calculations. In fact, the behavior of $1/T_1$ below T_c changes upon x, i.e. $1/T_1$ decreases less rapidly with increasing x as seen in Fig. 16, which is in contrast with the behavior for $x \leqq 0.15$ where the reduction rate just below T_c is almost the same as seen in Fig. 10. The effect of smearing of the density of states with increasing pair breaking associated with increasing x is considered to be reflected on the effective energy gap parameter, Δ/k_BT_c. Therefore, it should be noted that the obtained value of Δ/k_BT_c and its x-dependence may not be for real, but is phenomenological one. The importance is that both $1/T_1$ and K are explained by the same parameters.

Figure 25 shows N_{res}/N_o estimated from the NMR experimental results and T_c plotted against Sr content, x. In the compounds exceeding $x=0.15$, T_c decreases as N_{res}/N_o increases. As seen in Fig. 26, the tendency of decreasing of T_c/T_{co} upon N_{res}/N_o for LSCO is almost the same as the calculation based on the non-magnetic impurity scattering in the unitarity limit by Schmitt-Rink, Miyake, and Varma,[33] having not rapid reduction upon increasing N_{res}/N_o, although the change of the other parameters to affect T_c such as χ_Q, the hole content, and others are neglected. Here, ● and ○ represent the data for the LSCO and the Zn-doped $YBCO_7$,[12] respectively. Here T_{co} for LSCO and $YBCO_7$ are assumed as $T_{co}=38$ K for $x=0.15$ and 92 K, respectively, and N_{res} for both samples are assumed to be zero.

By the way, recently, Takagi *et al.*[9] have claimed that the superconductivity disappears for $x>0.15$ associated with the orthorhombic to tetragonal structural phase transition near $x=0.2$ and the bulk superconducting phase is confined to a much narrower composition region than believed.[10] However, it is not set-

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tled whether or not the orthorhombic and tetragonal phases exhibit superconductivity with lower-T_c as reported in the earlier works.[10] Our NMR and NQR results support that both phases (orthorhombic and tetragonal) are in the superconducting state from a microscopic point of view. The spin Knight shift extracted from the spectrum shifted as a whole below T_c and $1/T_1$'s for the NQR spectra with different structure such as the orthorhombic and the tetragonal as mentioned in §3.1 are reduced below T_c. We consider that the origin of the gapless superconductivity is due to the pair breaking by the inhomogeneity induced by the structural instability from the orthorhombic to tetragonal upon Sr doping. Thus it is expected that the pair breaking effect starts to suppress T_c above $x=0.12$.

§5. Conclusion

The spin dynamics and the static magnetic properties of $La_{2-x}Sr_xCuO_4$ were systematically investigated by measuring T_1 and Knight shift of ^{63}Cu. In the lightly-doped region, T_1 shows a T-independent behavior at low-T, which is ascribed to a spin diffusion effect from some small parts of the sample where a critical slowing down of the Cu spin fluctuations becomes appreciable, associated with a possible magnetic order around $x=0.12$ as seen in $La_{2-x}Ba_xCuO_4$.[29] The relaxation reflecting the structure of the superconducting order parameter is masked by this effect.

In the Sr content region exceeding $x=0.15$, there has been seemingly a doubt that the bulk superconductivity in $La_{2-x}Sr_xCuO_4$ exists in the narrower Sr content region than believed.[9,10] A systematic NMR study has, however, revealed that $1/T_1$ steeply decreases below T_c and a clear reduction of the spin Knight shift below T_c is deduced from the uniform shift of the NMR spectrum as a whole. From these results, it has been assured that there exist superconducting phases with T_c's lower than the maximum T_c of 38 K, even though the structural inhomogeneity is induced into the samples exceeding $x=0.12$. Furthermore, the observation of a $T_1T=$const. relation and of the residual spin Knight shift at low-T provides a clear signature that the superconductivity is of a gapless type. So far, the compounds with x more than 0.15 have been regarded to be in a over-doped region since T_c decreases. From the present systematic Cu NMR and NQR experiments together with the previous ^{17}O NMR experiment,[22] even the compound of $x=0.24$ with $T_c=18$ K does not attain the optimum hole content on the boundary between lightly- and over-doped region. The reduction of T_c is not due to the over-doping, but to the pair breaking associated with the structural inhomogeneity such as orthorhombic and tetragonal induced by the structural instability upon Sr doping. In this sense, it is remarkable that the samples exceeding $x=0.15$ in LSCO are not in the real heavily-doped regime like in the Tl(2201) and Tl(1212) systems. An interesting feature is that the gapless d-wave model, which was successfully applied to interpret the superconducting properties for the Zn doped $YBa_2Cu_3O_7$,[12] is useful in interpreting the T-dependence of T_1 and the Knight shift below T_c in the Sr content region exceeding $x=0.15$ as well.

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Note added in proof—Just sfter submitting this paper, we were aware of the paper by Y.-Q. Song *et al.* (Phys. Rev. Lett. **70** (1993) 3131). They have reported the T-dependence of Knight shift for $x=0.2$ in $La_{2-x}Sr_xCuO_4$. Their result is different from ours; namely, they concluded that the spin shift is zero at low-T, while we have shown that the spin shift remains finite associated with the pair breaking effect caused by some inhomogeneity in the sample.